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Muscle damage induced by maximal eccentric exercise of the elbow flexors after 3-week immobilization

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The present study investigated the effects of a 3-week immobilization (IM) on muscle damage induced by maximal eccentric exercise (MaxEC) to test the hypothesis that the IM would make muscles prone to muscle damage. Young healthy sedentary men were pseudo-randomly assigned to IM or control group (n = 12/group). Non-dominant arms of the IM group participants were immobilized at 90° elbow flexion by a cast for 21 days. All participants performed MaxEC consisting of five sets of six elbow flexor contractions by lowering a dumbbell set at 100% of pre-exercise maximal voluntary isometric contraction (MVC) strength of the non-dominant arm. This was performed at 2 days after the cast removal for the IM group. MVC torque, range of motion (ROM), muscle thickness (MT), muscle hardness, position sense (PS), and joint reaction angle (JRA) of the elbow flexors were measured at baseline, post-immobilization, and before, immediately after, and one to 5 days after MaxEC. The IM decreased MVC torque (−17 ± 2%), ROM (−2 ± 1%), MT (−7 ± 3%), and JRA (−12 ± 6%), and increased in muscle hardness (20 ± 6%) and PS (11 ± 2%) ($p < 0.05$). Changes in MVC (e.g., 2 days: −40 ± 5 vs. −30 ± 9%), ROM (2 days: −11 ± 2 vs. −9 ± 3%), muscle soreness (peak: 63 ± 22 vs. 48 ± 14 mm), plasma CK activity (peak: 7820 ± 4011 vs. 4980 ± 1363 IU/L), PS (maximal change: −23 ± 2 vs. −18 ± 3%), and JRA (maximal change: −37 ± 4 vs. −26 ± 3%) after MaxEC were greater ($p < 0.05$) for the IM than control group. These results supported the hypothesis and showed that the IM made the muscles more vulnerable to muscle damage induced by eccentric exercise.

KEYWORDS

creatine kinase activity, delayed onset muscle soreness, joint reaction angle, maximal voluntary contraction, muscle hardness, muscle thickness, position sense

1 | INTRODUCTION

Splints or casts are often applied to restrict movements of an injured bone, tendon and/or ligament, and to provide a proper alignment and reduce pain and swelling.[1] Following a period of immobilization, muscle strength and size, and proprioception decrease.[2] For example, Wall et al.[3] reported that as short as 5-day immobilization reduced maximal voluntary isometric contraction (MVC) torque of the knee extensors (−9 ± 2%) and quadriceps cross-sectional area (CSA: −4 ± 1%). A longer cast immobilization (e.g., 4 weeks) results in larger decreases in muscle strength of the knee extensors (−53 ± 9%) and flexors (−26 ± 13%), and quadriceps CSA (−21 ± 7%).[4] Valdes et at.[5] showed that elbow joint position sense was reduced 87% after 4-week immobilization by a sling applied for 8 h a day, indicating that proprioception was also affected by immobilization.

In order to restore muscles from such negative effects of immobilization, proper resistance exercise training is required after immobilization.[6] It has been documented that resistance exercise training focusing on eccentric (lengthening) contractions is effective for muscle strength gain and hypertrophy.[7] However, unaccustomed eccentric exercise induces muscle damage that is characterized by decreases in maximal voluntary contraction strength and joint range of motion, development of delayed onset muscle soreness (DOMS), swelling, increases in muscle stiffness or muscle hardness, and elevation of muscle proteins such as creatine kinase (CK) in the blood.[8] These symptoms could last more than a week, although the time course of recovery depends on the extent of muscle damage that is influenced by the intensity, volume, joint angle or muscle length of the eccentric exercise, and muscle groups used in the exercise.[8]

Prolonged muscle inactivity increases production of various inflammatory mediators such as reactive oxygen species (ROS) and superoxide anions, which are associated with reduced protein synthesis and increased muscle breakdown.[9] It is also known that mitochondria function decreases with inactivity, which could increase intracellular Ca^{2+} concentration.[9] It appears that type II (fast-twitch) muscle fibers are more prone to eccentric exercise-induced muscle damage than type I (slow-twitch) muscle fibers.[10] Since the magnitude of muscle atrophy induced by immobilization is greater for type II (−20%) than type I muscle fibers (−13%),[6] it is possible that immobilized muscles become more prone to muscle damage induced by high-intensity eccentric contractions. However, this has not been systematically investigated to the best of our knowledge.

Therefore, the present study investigated the effects of a 3-week arm immobilization on muscle damage induced by maximal eccentric exercise of the elbow flexors (MaxEC) in comparison to a control condition in which MaxEC was performed without immobilization. We hypothesized that the immobilization would exacerbate muscle damage indicated by changes in indirect markers of muscle damage and proprioception after MaxEC when compared with the control condition without immobilization.

2 | MATERIALS AND METHODS

2.1 | Participants and study design

Twenty-four sedentary healthy young men who had no musculoskeletal injuries of arms were recruited for this study. Each of them provided an informed consent to participate in the study that had been approved by the Research Ethic Committee of National Taiwan Normal University. The study was conducted in conformity with the policy statement regarding the use of human participants by the Declaration of Helsinki. All participants were asked and reminded to refrain from unaccustomed exercise or vigorous physical activity, maintain their normal diet, not to use their non-dominant arm as much as possible, and not to take any anti-inflammatory drugs or nutritional supplements (e.g., amino acid or protein) during the experimental period. These instructions were included in the information letter, and the participants were asked to inform the investigator if they were not able to follow the instructions.

The sample size was estimated using the data from our previous study in which changes in MVC strength of the elbow flexors after a 3-week of cast immobilization were compared between the immobilized and non-immobilized arms (Chen et al., unpublished). Based on the effect size of 1, it was estimated that at least 11 participants were necessary per group with an α level of 0.05, and a power $(1 - \beta)$ of 0.80.[11] Considering a possible estimation error, 12 participants were recruited for each group.

Maximal voluntary isometric contraction (MVC) torque of the elbow flexors was measured from the non-dominant arm after warm-up exercise and five practice trials. Based on the MVC torque, the participants were placed into one of the two groups (n = 12/group); control or immobilization (IM) group by matching the average baseline MVC torque between the groups as much as possible. The mean (±SD) age, height, body mass, and body mass index (BMI) of the 24 participants were 23.2 ± 2.1 years, 173.1 ± 5.0 cm, 69.5 ± 10.6 kg, and 23.2 ± 3.3 kg/m^2, respectively.

All participants in the IM group received immobilization of the non-dominant arm (opposite to the arm that could throw a ball better, faster, and further) for 3 weeks as explained below. The participants in the control group were asked not to change their normal daily activities during the 3-week period. All participants in the IM group performed maximal elbow flexor eccentric exercise by the immobilized arm at 2 days after the removal of the cast, while the participants in the control group performed the exercise after having the second baseline measures.

2.2 | Immobilization

The non-dominant arm of each IM group participant was immobilized by a cast at the elbow joint of 90° flexion and secured in a sling with a mild shoulder internal rotation to unload the elbow flexor muscles.[12] The participants were instructed to take off the sling only when taking a shower and sleeping, and the cast was not removed for 3 weeks. This protocol was adapted and modified from previous studies that showed 13–16% decreases in MVC strength of the elbow flexors after a 2-week immobilization of upper arm.[13,14]

2.3 | Maximal eccentric exercise (MaxEC)

A dumbbell was used for the eccentric exercise. To determine the dumbbell weight, MVC strength of the unilateral elbow flexors was measured by a loadcell (model DFG51; Omega Engineering, Stamford, CT) that was attached to a cuff surrounding the wrist of the exercise arm.[15] Each participant was seated on a custom-made preacher curl bench, placing the elbow joint angle at 90° (1.57 rad) and the shoulder joint angle at 45° (0.79 rad) flexion and 0° abduction.[15] The participant was instructed to flex the elbow joint maximally for 3 s, and this was repeated three times with a 45-s rest between attempts. The highest value of the three peak values was used to determine the dumbbell weight.[15]

All participants performed five sets of six eccentric contractions of the non-dominant elbow flexors, with a dumbbell corresponding to the 100% MVC strength of pre-exercise level. Since the dumbbell weight was close to or exceeded maximal strength at elbow extended angles, "maximal" effort was required for the participants to perform eccentric contractions. Thus, we considered the exercise as "maximal" although the 100% MVC load was not necessarily maximal for elbow flexed angles because of greater force generation capability in eccentric than isometric contraction. Each participant was instructed to lower a dumbbell from an elbow flexed (90° = 1.57 rad) to an elbow fully extended position (0° = 0 rad) in 3 s, and the investigator removed the dumbbell at the extended position, and the arm was returned to the start position without load. The eccentric contraction was repeated every 15 s, and a 2-min rest was given between sets.[15]

2.4 | Dependent variables

The dependent variables included MVC torque, elbow joint's range of motion (ROM), elbow flexor's muscle thickness (MT) by the ultrasound transverse images, upper arm circumference (CIR), muscle hardness (MH), delayed onset muscle soreness (DOMS), plasma CK activity, and position sense (PS) and joint reaction angle (JRA). The order of these measures was a blood draw for CK activity measure first, followed by muscle soreness, muscle hardness, CIR, ROM, MT, PS, JRA, and MVC on each time point. All measures were taken before and at 2 days post-immobilization for the IM group, and MVC torque and ROM were also measured within 30 min after the removal of the cast. The variables were measured from the non-dominant arm at baseline and 3 weeks later again for the control group. To assess muscle damage after MaxEC, all measures were assessed before, immediately after (except DOMS and plasma CK activity), and one, two, three, four, and 5 days after the exercise from the exercised arm.

The test–retest reliability of the dependent variables indicated by the coefficient of variation (CV) was demonstrated to be smaller than 9.9% in our previous studies performed in the same laboratory by the same investigators.[16,17]

2.4.1 | MVC torque

The MVC torque measurement were the same as that of our previous studies, thus the details can be found in elsewhere.[17] Briefly, the MVC torque was measured at 90° (1.57 rad) elbow flexion, where the full elbow extension angle was considered as 0° (0 rad), using a dynamometer (Biodex System S4, Biodex Medical Systems). Participants were verbally encouraged to maximally flex the elbow joint for 3 s, with a 45-s rest between three attempts. The peak value during the 3-s contraction was recorded, and the highest value of the three peak values was used for further analysis.[18–20]

2.4.2 | Range of motion (ROM)

The ROM of the elbow joint was determined as the difference between the elbow joint angles of maximal voluntarily flexion and extension measured by a manual goniometer.[15] Three measurements were taken for each angle, and the mean of the three measurements was used to calculate ROM.[15]

2.4.3 | Upper arm circumference

While each participant was standing, relaxing and letting the arm hang down by his side, the upper arm CIR was measured at the mid-portion of the upper arm, between the acromion process of the clavicle to the lateral epicondyle of the humerus, using a Gulick tape measure (Creative Health Products). The measurements were taken three times by the same examiner, and the mean of the three measures was used for statistical analysis.[15]

2.4.4 | Muscle thickness

This measure was adopted from our previous study,[17] and the details of the measurements can be found in the article. Briefly, B-Mode ultrasound images were obtained from the exercised upper arm by a Terason t3000 Ultrasound System (Terason Co.) with a 7.5-MHz linear probe. The probe was placed on the mid-portion of the upper arm for transverse images, while the participant was sitting on a chair with the forearm on an armrest. The gains and contrast were kept constant over the experiment period, and all images were saved to a computer (HP Workstation xw4400, HP xw4400 Workstation Base Unit, Singapore) and analyzed by a computer image analysis software (ULT File Reader for Windows, Broadsound Co.). A 7.5-MHz scanning head was placed on the measurement site without depressing the dermal surface. A combined biceps brachii and brachialis muscle thickness was determined as the distance from the subcutaneous adipose layer to the surface of the humerus bone.[17]

2.4.5 | Muscle hardness

Muscle hardness was measured by a tissue hardness algometer (OE-220, Ito Co. Ltd.) at the mid-portion of the biceps brachii (as the same site as that for the MT measure) when each participant lay supine with the testing arm was fully extended and relaxed on a message bed. Muscle hardness was measured three times on each time point, and the mean values of the three measures were used for further analysis.[21]

2.4.6 | Muscle soreness

Muscle soreness was quantified using a visual analog scale (VAS) that had a 100-mm continuous line with "not sore at all" on one side (0 mm) and "very, very sore" on the other side (100 mm). The investigator asked the participant to rate his perceived soreness on the VAS when the muscles were passively extended for the ROM (120°–0° of elbow flexion angles) measures.[15,22]

2.4.7 | Plasma CK activity

Approximately, 5-ml of venous blood was withdrawn by a standard venipuncture technique from the cubital fossa region of the arm and centrifuged for 10-minute to extract plasma, and plasma samples were stored at −80°C until analyses. Plasma CK activity was assayed spectrophotometrically by an automated clinical chemistry analyzer (Model 7080; Hitachi, Co. Ltd.) using a commercially available test kit (Roche Diagnostics).[23]

2.4.8 | Position sense (PS) and joint reaction angle to release (JRA)

PS and JRA measures were adopted from our previous studies,[16,24] and the details of the measurements can be found in the articles. Briefly, each participant seated and blindfolded for both PS and JRA testing, as the same position required as MVC testing on the isokinetic dynamometer. As for PS, the investigator positioned the testing arm of each participant to 45° (0.78 rad), held it for 10 s, and returned it passively to a full elbow extension (0° = 0 rad), then asked the participant to stop the elbow flexion movement at 20°/s by pressing a stop button by the hand of the opposite arm at the perceived target angle. For JRA test, the testing arm of the participant was passively moved by the investigator from 90° (0.79 rad) elbow flexion to a target angle (i.e., 45°) slowly (approximately 20°/s), then the testing arm was stopped and fully relaxed at a target angle for 5–10 s when it reached the target angle showing on the computer screen, then the investigator suddenly released the lever arm without warning. The participant was asked to stop the lever arm as soon as the investigator released the lever arm. The difference between the target angle and the actual angle (PS) or the angle at stopping (JRA) was recorded, respectively. The time between trials for the same test was 10 s, and 3 min of rest were provided between PS and JRA measurements. The difference from the target angle of the mean value of the two closest to the reference angle among four trials was used for further analysis.

2.5 | Statistical analyses

A Shapiro–Wilk test was used to check the normality assumption of the data, which demonstrated that normality assumption was met for all variables in the current study. Changes in all measures after the 3-week period from the baseline were assessed by a paired *t*-test for the IM and control groups, separately. Changes in the dependent variables after MaxEC were compared between the IM and control groups by a mixed-design of two-way ANOVA. When the ANOVA found a significant interaction effect, a Tukey's post hoc test was performed. Eta-squared values ($\eta 2$) were calculated as measures of effect size when necessary, and they were considered that a value of ~0.02 is a small effect; ~0.13, a medium effect; and >0.26, a large effect.[25] A significant level was set at $p \leq 0.05$. The data were presented as mean ± SD, unless otherwise stated.

3 | RESULTS

3.1 | Baseline measurements

No significant differences ($p > 0.05$) in any of the dependent variables were observed before the immobilization between the IM and control groups (Table 1).

3.2 | Effects of 3-week immobilization

The control group showed no significant ($p > 0.05$) changes in any variables over the 3-week period (Table 1). The IM group showed decreases ($p < 0.001$) in MVC torque (−43 ± 4%) and ROM (−12 ± 3%) from the baseline at ~30-minute after the removal of the cast. At 2 days post-IM, significant ($p < 0.001$) decreases in MVC torque (−17 ± 2%, $\eta^2 = 0.949$), ROM (−2 ± 1%, $\eta^2 = 0.840$), CIR (−4 ± 1%, $\eta^2 = 0.943$), MT (−7 ± 3%, $\eta^2 = 0.802$) and JRA (−12 ± 6%, $\eta^2 = 0.787$), and increases in muscle hardness (20 ± 6%, $\eta^2 = 0.930$) and PS (11 ± 2%, $\eta^2 = 0.974$) from the baseline were evident (Table 1).

3.3 | Exercise

The dumbbell used for MaxEC was heavier ($p = 0.006$) for the control (24.4 ± 2.8 kg) than the IM group (21.8 ± 1.8 kg). The total workload was also greater ($p = 0.006$) for the control (732 ± 85 kg) than the IM group (655 ± 55 kg).

3.4 | Changes in variables after MaxEC

Both groups showed significant ($p < 0.05$) changes in all variables after MaxEC (Figures 1–3). Since the baseline values before MaxEC were different between the IM and control groups due to the immobilization, relative changes to the pre-exercise values are shown in Figures 1 and 3. However, absolute changes are shown in Figure 2 (muscle soreness, CK activity), because their pre-exercise values were not significantly ($p > 0.05$) different. The ANOVA (interaction effect) showed that the changes were greater for the IM than the control group for MVC torque ($p < 0.001$, $\eta^2 = 0.356$), ROM ($p = 0.001$, $\eta^2 = 0.272$), CIR ($p < 0.001$, $\eta^2 = 0.534$), MT ($p < 0.001$, $\eta^2 = 0.418$), and muscle hardness ($p = 0.004$, $\eta^2 = 0.243$) as shown in Figure 1. Increases in muscle soreness ($p = 0.005$, $\eta^2 = 0.254$) and plasma CK activity ($p < 0.001$, $\eta^2 = 0.359$) were greater for the IM than the control group (Figure 2). Changes in PS ($\eta^2 = 0.489$) and JRA ($\eta^2 = 0.468$) were also greater ($p < 0.001$) for the IM than the control group (Figure 3).

TABLE 1 Changes (mean ± SD of 12 participants, their ranges in the brackets) in MVC torque of the elbow flexors, ROM, CIR, MT, MH, PS, and JRA of the elbow flexors before (Baseline) and 2 days after 3-week immobilization (Post) for the immobilized arm of the immobilization group and non-dominant arm of the control group

	Immobilization ($n = 12$)		**Control ($n = 12$)**	
	Baseline	**Post**	**Baseline**	**Post**
MVC (Nm)	49.7 ± 8.5 (38.7–65.0)	41.4 ± 6.7*# (32.5–51.8)	48.0 ± 3.8 (43.0–56.0)	47.1 ± 4.4 (42.0–55.1)
ROM (°)	141.4 ± 4.2 (133.0–146.0)	139.3 ± 4.6*# (131.0–145.0)	144.5 ± 3.5 (140.0–150.0)	144.3 ± 4.0 (1139.0–150.0)
CIR (mm)	298.5 ± 29.2 (260.0–355.3)	287.8 ± 29.5*# (245.7–346.7)	296.4 ± 13.8 (281.6–315.5)	296.4 ± 13.9 (281.5–315.7)
MT (mm)	3.41 ± 0.43 (2.72–4.05)	3.18 ± 0.43*# (2.45–3.81)	3.39 ± 0.24 (3.09–3.77)	3.40 ± 0.25 (3.08–3.75)
Hardness (N/cm^2)	33.6 ± 3.4 (27.6–39.2)	40.1 ± 3.5*# (33.3–44.4)	35.5 ± 4.1 (31.1–42.6)	35.2 ± 4.5 (31.0–43.0)
PS (°)	45.1 ± 1.9 (42.0–48.0)	50.1 ± 1.6*# (47.0–52.0)	45.8 ± 3.0 (42.0–53.0)	45.9 ± 2.6 (43.0–52.0)
JRA (°)	40.1 ± 1.2 (38.0–43.0)	35.4 ± 2.0*# (32.0–39.0)	39.8 ± 1.8 (37.0–42.0)	39.7 ± 0.9 (38.0–41.0)

Abbreviations: CIR, upper arm circumference; JRA, joint reaction angle to release; MH, muscle hardness; MT, muscle thickness; MVC, maximal voluntary isometric contraction; PS, position sense; ROM, range of motion.

*A significant ($p < 0.05$) difference from the baseline value.

#Significant ($p < 0.05$) difference from the control group.

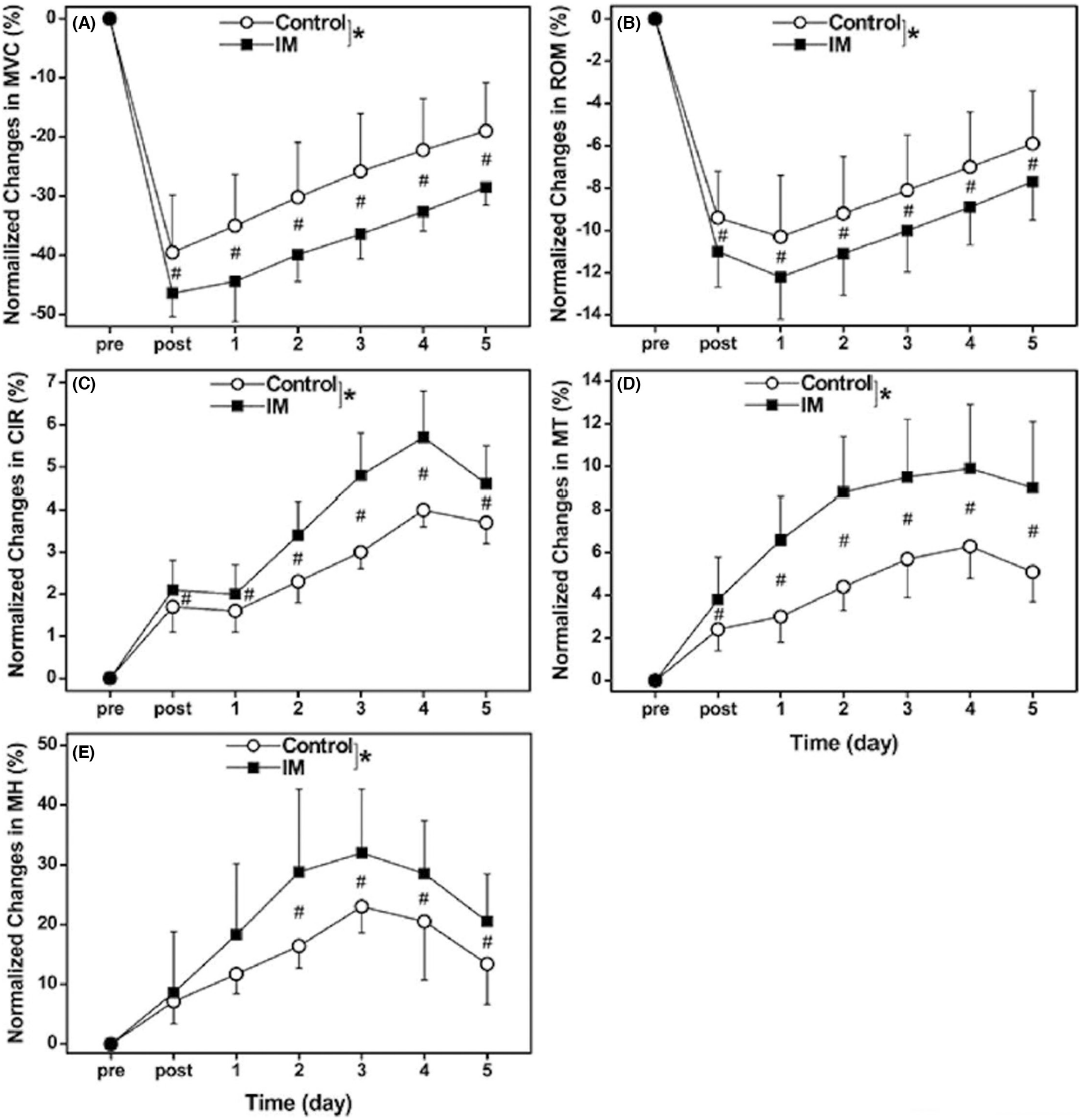

FIGURE 1 Normalized changes (mean ± SD) in maximal voluntary isometric contraction (MVC) torque (A), range of motion (ROM) (B), upper arm circumference (CIR) (C), muscle thickness (MT) (D), and muscle hardness (MH) (E) from the baseline (pre, 0%) at immediately after (post) and one, two, three, four, and 5 days after 30 maximal eccentric exercise of the non-dominant elbow flexors for the immobilization (IM) and control groups. * indicates a significant ($p<0.05$) interaction effect by mixed-design two-way ANOVA. # indicates a significant ($p<0.05$) difference between groups for each time point based on the post-hoc test.

4 | DISCUSSION

The present study found that the 3-week immobilization decreased muscle function, induced muscle atrophy, and impaired proprioception (Table 1), and that the magnitude of changes in the muscle damage markers after MaxEC was significantly greater for the IM than control group (Figures 1–3). These results appear to support the hypothesis that the immobilization would make the muscles more prone to muscle damage induced by eccentric exercise.

As shown in Table 1, the 3-week immobilization induced significant decreases in MVC torque (−17 ± 2%),

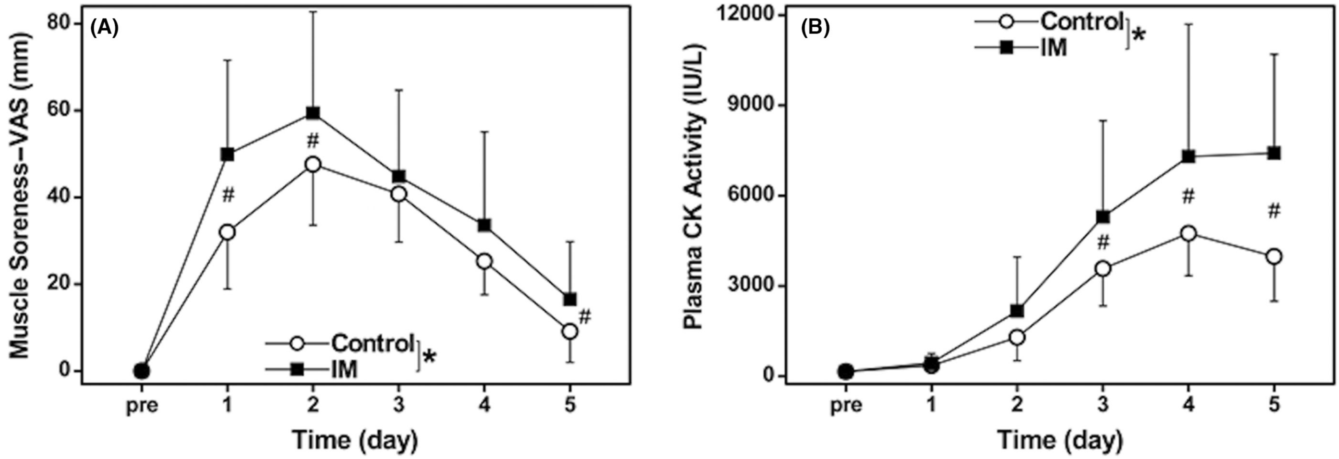

FIGURE 2 Changes (mean ± SD) in muscle soreness assessed by a 100-mm visual analog scale (VAS) (A) and plasma creatine kinase (CK) activity (B) before (pre), and one, two, three, four, and 5 days after 30 maximal eccentric exercise of the non-dominant elbow flexors for the immobilization (IM) and control groups. * indicates a significant ($p < 0.05$) interaction effect by mixed-design two-way ANOVA. # indicates a significant ($p < 0.05$) difference between groups for each time point based on the post-hoc test.

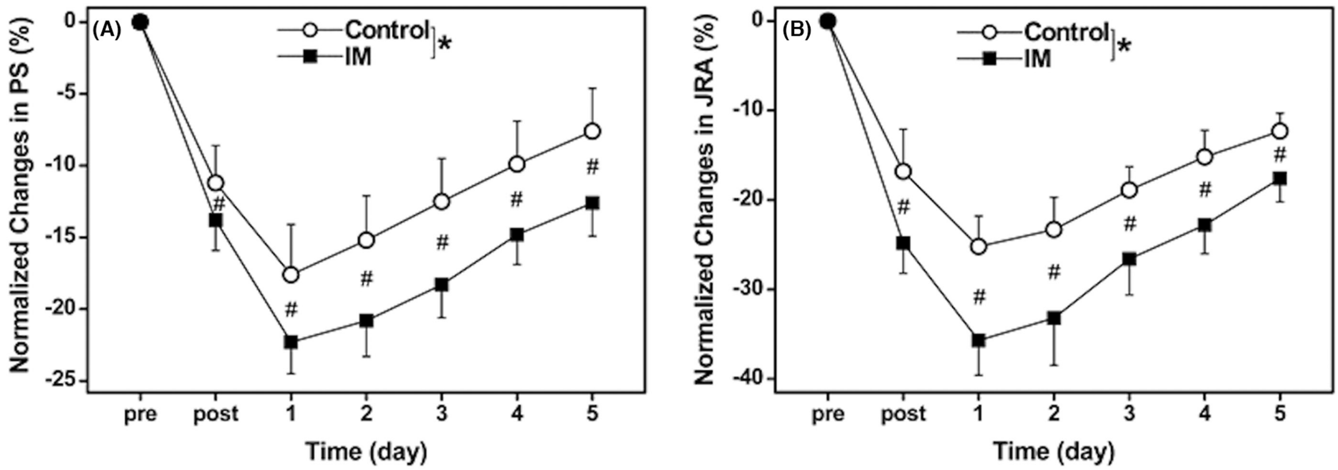

FIGURE 3 Normalized changes (mean ± SD) in position sense (PS) (A) and joint reaction angle to release (JRA) (B) of the elbow flexors from the baseline (pre, 0%) at immediately after (post) and one, two, three, four, and 5 days after 30 maximal eccentric exercise of the non-dominant elbow flexors for the immobilization (IM) and control groups. * indicates a significant ($p < 0.05$) interaction effect by mixed-design two-way ANOVA. # indicates a significant ($p < 0.05$) difference between groups for each time point based on the post-hoc test.

ROM (−2 ± 1%), MT (−7 ± 3%), and JRA (−12 ± 6%), and increases in muscle hardness (20 ± 6%) and PS (11 ± 2%) that were measured at 2 days after the removal of the cast. These results were in line with the findings of the previous studies in which the effects of immobilization on muscle atrophy and muscle function were examined.[5,26,27] For example, Pearce et al.[26] reported 6% decrease in muscle thickness and 20% decrease in one-repetition maximum (1-RM) strength of the elbow flexors after a 3-week sling immobilization applied for 15 h a day. Valdes et al.[5] showed that a 4-week sling immobilization applied 8 h a day decreased MVC isometric (−22%) and 1-RM strength (−14%), arm circumference (−5%), and position sense (−87%). Compared with the study by Valdes et al.[5] the changes in the position sense in the present study were smaller. This appears to be due to the difference in the method to assess the position sense such that Valdes et al.[5] assessed it as an average difference from a targeted angle at 30% and 50% of active ROM when the participant lied on a bed in a supine position, but the present study assessed it as a difference from the 45° of elbow flexion when the participant seated on a dynamometer. The increase in PS and decrease in JRA both indicate that proprioception was impaired by the immobilization.

To the best of our knowledge, this was the first study to report the effect of immobilization on muscle hardness. It is known that collagen synthesis decreases in response to joint immobilization[28] and that disuse increases breakdown of the extracellular matrix (ECM) particularly in the perimysium.[29] It is also shown that titin is reduced by inactivity.[30] These may have contributed to the increased muscle hardness. Thus, the eccentric exercise was performed by weaker and stiffer muscles by the participants in the IM than the control group. It should be noted that the dumbbell weight used for the MaxEC was proportional to the MVC torque, thus 11% lighter for the IM than the control group, and the total workload was also 11% smaller for the IM than the control group.

The changes in the muscle damage and proprioception markers following MaxEC in the control group were similar to those following the same eccentric exercise of the unilateral elbow flexors performed by sedentary young men without immobilization observed in our previous study[31] in which the same eccentric exercise as that of the present study was performed by 15 participants whose physical characteristics were also similar to those of the present study. The average changes in MVC torque at 2 days post-exercise was $-33\pm11\%$, ROM at 2 days post-exercise was $-10\pm3\%$, increases in CIR at 4 days post-exercise was $5\pm1\%$, peak muscle soreness was 50 ± 31 mm and peak plasma CK activity was 5958 ± 607 IU/L. These were comparable to the changes observed for the control group for the same time point (MVC torque: $-30\pm9\%$, ROM: $-9\pm3\%$, CIR: $4\pm1\%$) or peak values (muscle soreness: 52 ± 9 mm, plasma CK activity: 4980 ± 1363 IU/L) in the present study. Therefore, it is most likely that the responses to the MaxEC in the control group shown in Figures 1–3 represent typical changes in the variables after an acute bout of MaxEC performed by non-resistance-trained young men.

If no effects of the 3-week immobilization had existed, the changes in the variables after MaxEC in the IM group should have been similar to those shown by the control group. However, the changes in all variables after MaxEC in the IM group were significantly greater than those of the control group (Figures 1–3). The magnitude of decrease was 9% (1 day post-exercise) – 10% greater (5 days post-exercise) for MVC torque, and 15%–23% greater for ROM in the IM than the control group. Peak muscle soreness was 30% greater and peak plasma CK activity was 39% greater for the IM than the control group. Regarding the proprioception markers, PS and JRA showed 6% and 11% greater changes for the IM than the control group. These indicate that muscle damage induced by MaxEC was exacerbated by the 3-week immobilization.

This was the first study to demonstrate that muscles become more prone to muscle damage induced by eccentric exercise after immobilization. It is likely that type II muscle fibers are preferentially damaged by eccentric exercise,[10] and type II muscle fibers are more affected by immobilization.[6] Thus, it seems reasonable to assume that the immobilization makes type II muscle fibers more susceptible to eccentric exercise-induced muscle damage. It has been well documented that muscles become more immune to eccentric exercise-induced muscle damage by the repeated bout effect[8,32] and performing preconditioning exercise such as low-intensity eccentric contractions[33] or maximal isometric contractions at a long muscle length.[34] Thus, muscles appear to develop protective mechanisms rapidly against eccentric exercise-induced muscle damage. In the present study, the immobilization period was 3 weeks, but it is interesting to investigate whether a shorter period of immobilization still affects the muscle damage. The preconditioning effect induced by low-intensity eccentric contractions[33] or maximal isometric contractions at a long muscle length[34] were shown to last 14 and 4 days, respectively. This suggests that the adaptations obtained by such contractions disappear within 3 weeks. It is interesting to investigate whether the repeated bout effect or preconditioning effect is affected by immobilization. It may be that the immobilization after eccentric exercise or isometric exercise with a long muscle length attenuate the protective effect.

Prolonged muscle inactivity increases production of various inflammatory mediators such as ROS and superoxide anions, which could reduce muscle protein synthesis and increase muscle protein breakdown.[9] It has been reported that limb immobilization for only 4 days already causes a decline in functional strength in older adults.[35] At the cellular level, as short as 48 h of immobilization of one leg significantly increased messenger ribonucleic acid (mRNA) content for components of the ubiquitin-proteasome pathway (UPP) and metallothionein function while decreasing mRNA and protein for extracellular matrix (ECM) components as well as decreased phosphorylation of Akt or protein kinase B.[36] In the present study, MVC torque and muscle thickness were reduced 17% and 7%, respectively post-immobilization (Table 1). In a review paper, Herbert[37] proposed that immobilization in a shortened muscle position, which was the case for the elbow flexors in the present study, reduced sarcomere number in series but an increase in the intramuscular connective tissue. Previous studies reported that less compliant and/or stiffer muscles are more susceptible to eccentric exercise-induced muscle damage.[38] McHugh et al.[38] showed that the magnitude of muscle damage induced by maximal isokinetic eccentric contractions of the knee flexors was greater for stiffer than compliant hamstrings. Thus, it is possible the increased muscle hardness by immobilization made the elbow flexors less resilient to mechanical strain in MaxEC. Regarding the impaired proprioception, the errors of PS and JRA of the immobilized arm increased (Table 1). The impairment of PS and JRA may be associated with the muscle and tendon dysfunction, which could exacerbate muscle damage. These may explain why the muscle damage was greater in the IM than the control group.

To recover from muscle atrophy and strength loss induced by immobilization, resistance exercise training is necessary. In the resistance exercise, eccentric contractions are normally included. Thus, it is possible that the post-immobilization exercise induces severe muscle damage. This should be noted by practitioners when they prescribe and perform exercise rehabilitation program. It has been reported that resistance exercise including eccentric contractions of non-immobilized limb is effective for attenuating muscle atrophy and strength loss of the immobilized limb via the cross-education effect.[5,27] Valdes et al.[5] reported that eccentric resistance training performed by the non-immobilized arm three times a week during the 4-week sling immobilization (8 h/day) prevented decreases in MVC torque and arm circumference. Thus, it appears that eccentric contractions of the non-immobilized limb provide better protection against muscle atrophy and loss of muscle function in immobilization.

Several limitations exist in the present study. First, only young sedentary healthy men were used in the study, hence it is not known whether the results of the present study are applicable for other populations such as women, middle- and old-aged populations, people with severe or limb bone fracture. Second, it is well documented that the responses to eccentric exercise are different between arm and leg muscles,[39,40] but the present study focused on the elbow flexors. Thus, the present study results cannot be generalized to other muscles (e.g., knee extensors, plantar flexors). Third, the present study did not monitor physical activity during the immobilization and nutrition intake of the participants during the experiment, although they were instructed to not to perform any exercise and take any nutrition supplements during the experiment. Lastly, the present study was rather descriptive, and did not investigate the potential mechanisms underpinning the negative effects of immobilization on muscle damage.

In conclusion, the present study showed that the 3-week immobilization made the elbow flexor muscles more vulnerable to muscle damage induced by maximal eccentric exercise performed at 2 days after removal of the cast. More studies are warranted to investigate the underpinning mechanisms how immobilization exacerbated the extent of muscle damage and how to minimize muscle damage in immobilized muscles.

5 | PERSPECTIVE

The current study demonstrated that the 3-week immobilization of the elbow flexors resulting in decreased muscle function, size, and proprioception, and increased muscle hardness, increased the magnitude of muscle damage induced by elbow flexor eccentric exercise that was performed at 2 days after the removal of the cast. After immobilization, resistance exercise training is required to restore the decreased muscle function and muscle size. It is important to note that muscles become more prone to muscle damage after immobilization (Figures 1–3). Thus, to attenuate or prevent muscle damage after immobilization, training should be started from low intensity and low volume, even if high-intensity and large volume eccentric contractions had been performed before immobilization. Before including eccentric contractions, it may be better to start with isometric contractions at a long muscle length,[34] then eccentric contractions should be re-introduced from low intensity. It is also better to do eccentric training of the non-immobilized limb[5,27] during immobilization to avoid strength loss and induced muscle atrophy of the immobilized limb by maximizing the cross-education effect. Future study is warranted to establish the most effective strategies to counteract the negative immobilization effects.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request. The data are not publicly available due to privacy or ethical restrictions.

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